

Reflection Journal: Stroke Prediction Machine Learning Project

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Course: ITAI1371 - Machine Learning Final Project

1. Key Technical Learnings

Implementing this end-to-end classification pipeline provided critical hands-on insights into the complexities of real-world datasets:

The Danger of Raw Accuracy:

Initially, working with the stroke prediction dataset highlighted how misleading raw accuracy can be. With a 95% majority class imbalance, a naive model achieving 95% accuracy is actually failing completely at its core objective predicting strokes. Shifting our evaluation focus to F1-Score, Recall, and ROC-AUC was vital for measuring true predictive utility.

Mastering SMOTE:

Learning how to implement the Synthetic Minority Over-sampling Technique (SMOTE) demonstrated how synthetic data generation can help models learn the decision boundaries of minority classes.

Preventing Data Leakage:

Perhaps the most crucial lesson learned from professor feedback was the strict separation of data processing phases. Applying SMOTE exclusively to the training set via an `ImbPipeline` while leaving the validation and test sets entirely untouched ensured our evaluation framework remained honest, uncorrupted, and reflective of unseen real-world data.

2. Model Comparison Insights

Comparing six distinct classifiers alongside ensemble models revealed distinct algorithmic behaviors:

Linear vs. Tree-Based Models:

Logistic Regression and Support Vector Machines performed decently with linear separations, but ensemble methods like Random Forest and Gradient Boosting captured complex, non-linear feature interactions (such as the combined effect of age, average glucose levels, and hypertension) much more effectively.

The Power of Ensembles:

Combining top-performing models through an Averaged Probability Ensemble and a Bayesian Ensemble helped smooth out individual model variance, yielding robust and stable ROC-AUC scores across both validation and test sets.

3. Conclusion & Future Growth

This project bridged theoretical machine learning concepts with practical engineering challenges. Building a complete pipeline from exploratory data analysis and preprocessing to rigorous evaluation and GitHub deployment greatly strengthened my confidence in designing robust classification systems for medical and clinical datasets.